Complete-criterion computation for penalized least-squares trend selection: a manifest-audited Windows timing study

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Abstract

Selecting a smoothing parameter for finite-difference penalized least squares requires computation of a complete statistical criterion, not merely a fitted trend. We audit implementations of ordinary cross-validation (CV), generalized cross-validation (GCV), corrected Akaike information criterion and Gaussian marginal likelihood on common normalized-smoothness candidate grids. A focused 29-case campaign has 97 additional seed trials per case on a Windows workstation. An audited six-stage cumulative report confirms 2813 scheduled case-seed jobs, including 57 nonzero process exits. Manifest-driven ingestion explicitly preserves incomplete and numerically invalid searches, resource exclusions and distinct worker-contention phases. Of 92,679 recorded or explicitly masked timing rows, 76,220 satisfy complete-search status and candidate coverage; selected-criterion agreement remains a double-precision parity check rather than independent high-precision certification. For $\boldsymbol{n = 256}$, order-three conditional CV, a matched banded/SVD cold-time ratio reverses from 0.70 (95% seed-bootstrap interval 0.67–0.72) for a single candidate and series to 4.85 (4.57–5.12) for a 100-candidate search. Many larger order-four full grids fail despite successful single-candidate evaluations. These observations show that setup amortization, selection criterion, candidate mapping, workload, numerical admissibility and execution load jointly determine measured costs. No universally optimal solver or improvement in forecasting accuracy follows.

Keywords: penalized least squares, Whittaker–Henderson smoothing, cross-validation, smoothing-parameter selection, reproducible timing, numerical reliability

1 Introduction

Finite-difference penalized least squares (PLS) is an established trend-smoothing technique Whittaker (1923); Guerrero (2007). Selecting its smoothing parameter by cross-validation (CV), generalized cross-validation (GCV), corrected Akaike information criterion (AICc), BIC, or a Gaussian marginal-likelihood criterion entails evaluating complete statistical scores, rather than merely obtaining a fitted trend Cortés-Toto et al. (2017). Efficient banded matrix algorithms and related Whittaker–Henderson smoothing procedures have been developed for decades Weinert (2007, 2012); Biessy (2026); this article does not claim their invention.

We ask when a one-time dense spectral decomposition can amortize its cost over repeated full-criterion evaluations, and when a banded procedure remains preferable. The answer depends jointly on sample length, difference order, criterion, smoothness-candidate count, number of series, numerical admissibility, and cache reuse. In particular, observationwise leverage required by ordinary CV differs from the scalar trace needed for GCV. Comparison with an explicit dense-inverse reference and a sparse-Cholesky/banded-trace hybrid is possible only where both the full search and its numerical checks succeed.

The principal methodological contribution of this article is a manifest-driven audit of completed empirical results. An experiment may report every scheduled job as recorded while retaining unsuccessful processes, partial searches and interrupted attempts. We recover the authoritative original and continuation runs, preserve error statuses, and estimate matched speedups with seed-cluster rather than rowwise uncertainty. This is an empirical study of complete-criterion implementation costs, separate from the repository's computational-complexity, high-precision numerical and NGSIM application manuscripts.

2 Statistical and computational framework

2.1 Established estimator and normalized smoothness

Let $y \in \mathbb{R}^n$ be equally spaced observations, let $D_d \in \mathbb{R}^{(n-d)\times n}$ denote the order-d forward-difference operator, $1 \le d < n$, and let μ be a specified mean of the penalized differences. The existing PLS estimator is

$$\widehat{\tau}_{\lambda,\mu} = \arg\min_{\tau\in\mathbb{R}^n} \left\{ \|y-\tau\|_2^2 + \lambda \|D_d\tau - \mu\mathbf{1}\|_2^2 \right\}, \qquad \lambda \ge 0. \tag{1}$$

If $D_d p = \mu\mathbf{1}$, then

$$\widehat{\tau}_{\lambda,\mu} = p + H_\lambda(y-p), \qquad H_\lambda = (I + \lambda D_d^\top D_d)^{-1}. \tag{2}$$

The difference matrix has full row rank in exact arithmetic, with $n-d$ positive penalty eigenvalues δ_j. Normalized trace smoothness is the monotone mapping Guerrero (2008)

$$S(\lambda) = \frac{n - \operatorname{tr}(H_\lambda)}{n-d} = \frac{1}{n-d}\sum_{j=1}^{n-d}\frac{\lambda\delta_j}{1+\lambda\delta_j}, \qquad S(0)=0,\ S(\infty)=1. \tag{3}$$

Thus S is a coordinate for the *same* PLS estimators and selection objectives, not a new penalty. The exact $S=1$ endpoint represents the constrained limiting polynomial and is not approximated by an enormous floating-point penalty.

2.2 Complete statistical criteria

Let $r_\lambda = y - \widehat{\tau}_{\lambda,\mu}$ and $h_{ii} = (H_\lambda)_{ii}$. For fixed numeric μ, classical CV and GCV take the forms

$$\mathrm{CV}(\lambda) = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{r_{\lambda,i}}{1-h_{ii}}\right)^2, \tag{4}$$

$$\mathrm{GCV}(\lambda) = \frac{\|r_\lambda\|_2^2/n}{\{1-\operatorname{tr}(H_\lambda)/n\}^2}. \tag{5}$$

The original, versioned Experiment006 implementation defines finite-precision AICc, BIC and Gaussian marginal likelihood and their endpoint behavior. The observed-difference reference mean $\mu = \overline{D_d y}$ is a conditional full-sample plug-in: the reported CV is *not* the exact leave-one-out procedure that reestimates μ in every fold. Marginal-likelihood scores from different numerical formulas may differ by a λ-independent additive normalization; agreement of their minimizing S is checked without requiring identical absolute score levels. Undefined criteria and partial candidate evaluations cannot be timing winners.

Four workloads are measured: one candidate/one series, M candidates/one series, one candidate/R series, and M candidates/R series. Both direct-S mapping and once-mapped-λ policies evaluate the same candidates $S = j/M$ for $j = 1, \ldots, M$, with the endpoint $S=1$ handled analytically. Single-candidate tasks use the predeclared fixed smoothness. A valid complete-search time accounts for solver setup, smoothness-to-penalty translation and every candidate's criterion evaluation. The evaluator reports cold and warm cost accounting; a warm total is not proof of an independently spawned second process. At small n, both backends may pay a common spectral-mapper setup charge.

2.3 Four complete-criterion implementation families

The original benchmark contains full spectral SVD, banded Cholesky with selected-inverse criterion diagnostics, naive explicit dense inversion, and an optional sparse-Cholesky *plus separately charged banded diagnostics* hybrid. The hybrid is not a pure sparse Cholesky benchmark. All computations use one BLAS thread per job. Under fixed small difference order, a banded factorization has a nominal $O(nd^2)$ arithmetic

cost, while full dense spectral decomposition incurs asymptotically greater setup time and storage. Neither expression is measured wall time. The full search additionally incurs criterion-dependent residual, leverage or trace calculations, mapping and cache costs. Generalized operation counts and their proofs belong to the separate complexity companion; this paper reports seconds only where measured.

3 Experiment011 design and canonical ingestion

3.1 Predeclared case-seed campaign

The study is the focused 29-case Experiment011 design, not the unrelated 252,000-cell historical factorial *plan*. For each case, 97 additional seeded jobs use base seeds $10^9 + 10^6 i$ for $i = 0, \ldots, 96$, stored under indices $i + 3$. Design case identifiers fix n, d, criterion, amplitude, reference-mean policy, candidate count, method list, repeats, and the four workloads. Most non-smoke cases contain two within-seed repeats and rotated method ordering; the rotating block label is not itself an independent seed.

The final cumulative report records 2813 distinct case-seed jobs from the self-hosted Windows X64 runner PCAG103A-08. The controller used one BLAS thread per job but changed maximum concurrent workers during the campaign: 2231 jobs were recorded in the 24-worker phase, 52 in a one-worker phase, and 530 in a 12-worker phase. Phase limits are *not* actual instantaneous CPU occupancy. These phases are not pooled to claim isolated single-process latency, nor do summed per-method times establish campaign throughput.

3.2 Exact run selection and numerical masks

The final report depends on six successive cumulative checkpoints. Their predecessor hashes and inherited arrays were verified, yielding 2272 canonical records from the original run tag, then 11, 55, zero, zero and 475 additional records from successive continuations. Run tags are taken from the stage that first appended a record; selecting every directory recursively would incorrectly treat interrupted attempts as independent replications. The original report, source file, case and QA manifests, environment JSON and raw CSVs remain read-only. Each selected file is compared against its SHA-256 inventory entry and a verified local backup copy.

A row is classified as a complete status-admissible search only if the process completed, its saved status is successful, its time and selected score are finite, and the observed number of criterion evaluations equals MR. Paired methods must additionally agree on selected S within 10^{-10} and non-likelihood selected-score relative error within 10^{-7}, using the same case, seed, repeat, workload, coordinate, executable hash, machine environment and worker phase. Failed mappings, resource exclusions, unsuccessful processes, incomplete candidate searches and absent files remain explicitly masked. These tests establish recorded floating-point parity, *not* uniform independent high-precision correctness at large d.

Table 1 Valid matched complete-search banded/SVD ratios for a 100-candidate, one-series grid in the 24-worker phase. Data denotes the conditional observed-difference mean, versus the default fixed zero mean. Hybrid labels refer to the broader comparison design, not the SVD/banded pairing itself. Confidence limits resample independent seed clusters; numerical agreement is float64 selected-score parity only.

Criterion	Case design	n	d	seeds	ratio	95% CI
CV	pilot	64	1	96	2.34	[2.29, 2.39]
CV	smoke	64	3	97	4.03	[3.95, 4.11]
CV	pilot	128	3	97	5.40	[5.24, 5.55]
CV	pilot-data	128	3	97	5.56	[5.46, 5.65]
CV	confirm-data	256	3	97	4.85	[4.57, 5.12]
GCV	pilot	64	1	96	2.33	[2.28, 2.37]
GCV	pilot-hybrid	64	1	97	2.37	[2.31, 2.44]
GCV	smoke	64	1	97	2.40	[2.34, 2.46]
GCV	pilot-data	128	3	97	5.12	[5.01, 5.23]
GCV	pilot	128	3	97	4.99	[4.87, 5.10]

3.3 Paired seed-cluster uncertainty

For paired complete cold times $T_A, T_B > 0$, the speedup of method A relative to method B is $R = T_B/T_A$. Our summary takes the arithmetic mean of $\log R$ within each case-seed cluster, preserving repeated method/workload outcomes together, then exponentiates the across-seed mean. We estimate percentile 95% intervals using 700 deterministic bootstrap resamples of independent seed indices. No interval is reported for fewer than three independently seeded clusters. The intervals quantify within-design seed variation under recorded machine conditions, not uncertainty under different CPUs, load, OS installations, or changes in optimization software. Conditioning on successful paired searches can introduce survivor bias.

4 Results

4.1 Reported completion and measured admissibility

The completed campaign has 2813 unique scheduled case-seed jobs. However, 57 processes exited nonzero. The manifest-driven sparse timing tensor has 92679 measured or explicitly missing rows, of which 76220 (82.24%) satisfy complete-search status and candidate-coverage requirements. The other 16,459 rows include 11,208 numerical-failure statuses, 4224 mapping-failure statuses, 970 resource-budget exclusions and 57 jobs without necessary input files. The file inventory records 60 missing individual files across those failed jobs. The completed schedule is therefore *not* a full numerical success certificate.

The origin report and its verified same-disk backup have identical SHA-256 digests. The six-stage cumulative lineage is archived in the derived analysis summary and each accepted record is linked to its originating run folder. All original inputs remain

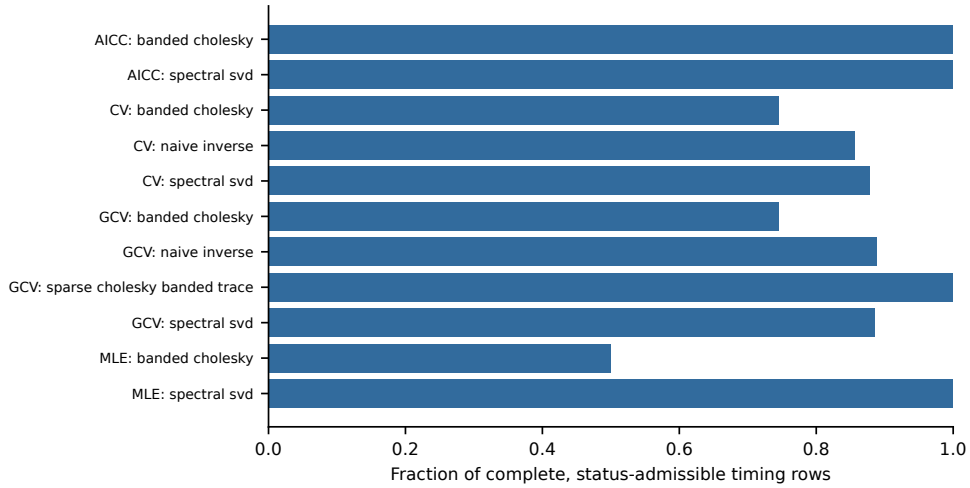


Fig. 1 Fraction of complete, status-admissible measured rows by criterion and method. Denominators retain failed, missing and resource-excluded cells. Method availability and difficulty vary by case, so these descriptive rates are not randomized failure-rate comparisons.

untouched. The backup improves protection against accidental edits but not loss of the Windows disk.

The complete-search distinction is material: for recorded $n = 1024, 2048, 4096$, $d = 4$ CV/GCV confirmation cases, single-candidate evaluations can complete even when the entire 100-candidate grid fails. A successful one-candidate timing is *not* promoted to an observed full-grid winner. Numerical/mapping issues are encountered by more than one backend and must not automatically be attributed to a single solver.

4.2 Criterion-specific matched costs

The most informative supported comparison is between spectral SVD and banded Cholesky on the same candidate grid. We define the table's reported ratio as banded-Cholesky complete cold time divided by spectral-SVD complete cold time. A ratio exceeding one favors spectral SVD; a ratio below one favors banded Cholesky. Table 1 lists selected valid one-series grid comparisons; Fig. 2 displays observed CV/GCV slices with independent-seed uncertainty.

For $n = 256$, $d = 3$, full-sample observed-difference μ, and conditional CV, we have 97 independent seed clusters in the 24-worker phase. The banded/SVD geometric cold-time ratio equals 0.698 (95% interval 0.671–0.724) for one candidate/one series; the corresponding 100-candidate grid/one-series ratio is 4.846 (4.567–5.125), and the 100-candidate grid/three-series ratio is 3.655 (3.450–3.860). The observed reversal reflects amortization and the complete criterion. It must not be generalized to an unmeasured (n, d, M, R) region or interpreted as contention-free latency.

For the $n = 64$, $d = 1$, $M = 100$ GCV pilot, complete one-series grid selection yielded a matched ratio of 2.328 (2.282–2.374) across 96 seed clusters. The same case's

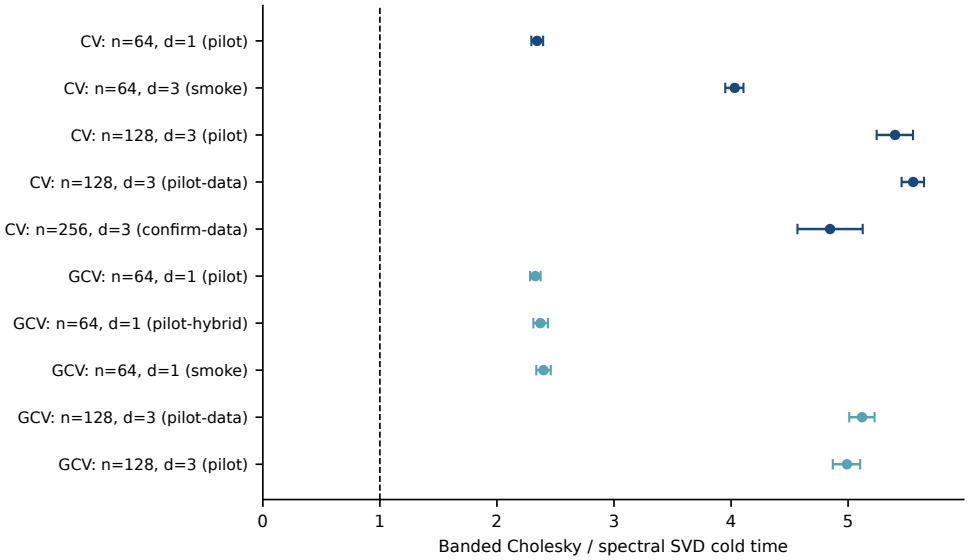


Fig. 2 Paired banded/SVD full cold-time ratios for available admissible CV and GCV 100-candidate, one-series direct-S cases. Bars represent seed-cluster 95% bootstrap intervals within the labeled execution phase, not independent per-row intervals. Missing large-n grids are deliberately omitted.

one-candidate/one-series ratio was 1.001 (0.982–1.018); an apparently decisive grid advantage can coexist with negligible isolated-candidate differences. Thus the number of criterion candidates is not a cosmetic experimental parameter.

4.3 Coordinate costs and workloads

Direct-S versus mapped-λ compares computation of *the same* smoothing candidates. In the $n = 256$, $d = 3$, conditional CV, $M = 100$, $R = 3$ spectral SVD case, the paired mapped/direct complete cold-time ratio is 0.580 (95% seed-bootstrap interval 0.570–0.590) in the 24-worker phase. This is evidence that caching the coordinate translation is useful for that particular repeated-grid workload; it is not a universal assertion about smoothness coordinates.

The underlying `timing_cell_summary.csv` retains medians and 10th–90th percentiles for setup, mapping, candidate evaluation, total cold time and total warm time in each phase. The four workloads in Fig. 4 are repeated measurements within a case-seed, not four statistically independent replications. No fit-only time is silently substituted for complete CV or GCV search time.

4.4 Reliability and sensitivity to exclusions

Figure 1 and the reliability CSV record actual denominators. In the 24-worker phase, banded-Cholesky CV has 10,848 valid complete rows out of 13,176 recorded rows (82.3%), compared with 13,176/13,176 for spectral CV. For GCV the reported

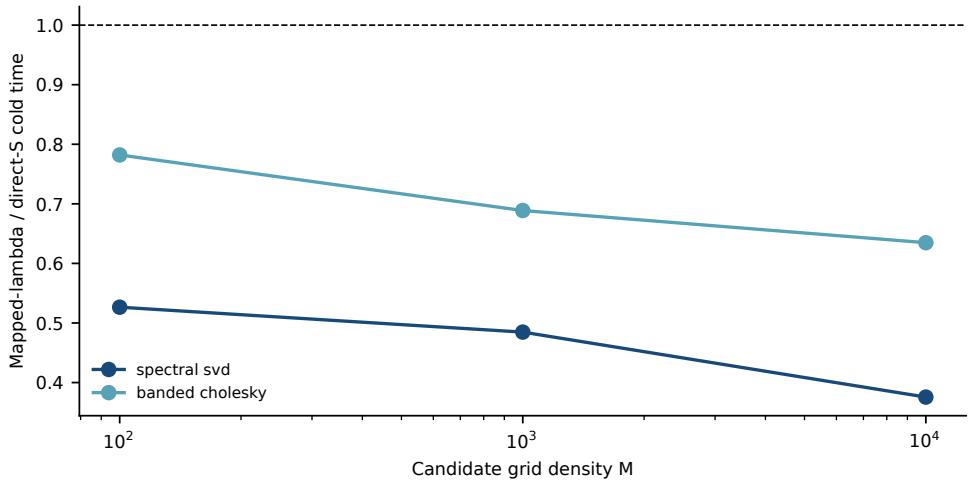


Fig. 3 Cell-level medians of geometric mapped-λ/direct-S cold-time ratios from numerically eligible grid/multiple-series pairs in the 24-worker phase. Values below one favor mapping once. A plotted point summarizes available design slices, not a randomized causal treatment.

fractions are 13,176/16,280 (80.9%) and 15,892/16,086 (98.8%), respectively. These aggregate rows do *not* share identical case denominators because resource gates and case method coverage differ. They describe the observed experiment, not a universal pairwise solver reliability probability. The full uncertainty and failure inventory remains available rather than being hidden by the valid-pair subset.

A series of large-order grids exhibits shared mapping failures and incomplete numerical outcomes, while smaller supported CV and GCV searches give valid paired comparison intervals. Consequently, comparisons based only on successful small grids may overstate the general reliability of either backend. The naive inverse is a reference only where an identical, valid measured pair exists; when dense memory or runtime budgets make it ineligible, no surrogate baseline speedup is invented. The optional sparse-Cholesky implementation is an explicitly diagnostic-charged hybrid and cannot be presented as a standalone sparse solver.

5 Discussion

The principal supported result is conditional, not a global ranking: depending on criterion and candidate count, the same banded and dense spectral implementations reverse their timing preference. A full grid rewards amortized spectral setup, but individual candidates can favor structured banded methods. The numerical status of the *entire* grid is essential to this comparison, especially at large n and high d.

The study has four important limitations. First, this was a single Windows host with 24-, one- and 12-worker execution phases, one BLAS thread each, and no complete occupancy instrumentation. The phase count is a concurrency limit, not a constant level of CPU contention, and time comparisons are not transportable estimates of

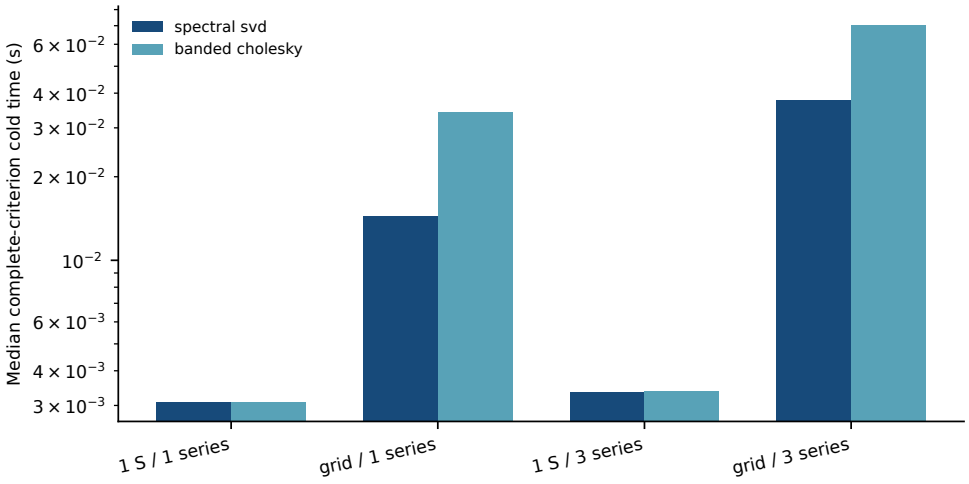


Fig. 4 Median complete-criterion cold times for one candidate/one series, grid/one series, one candidate/three series and grid/three series in the $n = 64$, $d = 1$, GCV, $M = 100$, direct-S pilot under the 24-worker phase. All times include recorded setup, mapping and candidate evaluation costs.

isolated latency. Second, excluding invalid complete searches creates potential survivor bias in otherwise matched speedups. Third, selected-criterion float64 parity does not independently certify forward errors or numerical behavior at all smoothing parameters. Fourth, measured criterion timing is not a measure of trend-estimation RMSE, predictive performance or physical jerk quality. High-precision numerical theory, operation-count complexity and the NGSIM vehicle application belong to separate companion work.

An informative follow-up would randomize method ordering and predeclare a smaller, numerically feasible high-order grid on an otherwise idle workstation, record processor occupancy and validate objective values against independent high-precision arithmetic. That study has not been performed here and no claims about its potential results are made.

6 Reproduction and data availability

The scripts and frozen *derived* results are archived in the public repository https://github.com/heri-espino/trendestimation, in the `paper/benchmarks/experiment011/pc08_empirical` directory. GitHub Actions run `42636962129` and its permanent results branch contain the source-to-file acceptance manifest, report-chain SHA-256 hashes, compressed sparse timing tensor, status/reliability tables, matched speedups, seed-bootstrap intervals, reproducible figure code, vector PDFs, PNG previews and claim-to-data ledger. The original final cumulative report's SHA-256 is

`c5d4898c83c731ada23a10957f80adc43616200206dc76573f365d38ba37f723`

The underlying *raw* measurements are preserved in an author-owned Windows checkout and a verified same-disk backup; they were not automatically pushed to GitHub or falsely represented as public data. Raw-data access and redistribution are subject to author approval.

The manifest-driven read-only reproduction command from the repository root is

```
python paper/benchmarks/experiment011/pc08_empirical/analyze_campaign.py
--original ORIGINAL --backup BACKUP --out NEW_OUTPUT
```

It does not rerun the numerical experiment. The official `sn-jnl` manuscript is compiled using the existing `trendestimation` Conda environment and `scripts/build_papers_pc08.py --paper springer`. The final build artifact and digest must be checked independently of the archived prior manuscript PDFs.

7 Conclusions

An audited schedule of 2813 case-seed jobs supplies many useful complete-criterion timing observations, but also demonstrates why numerical failures and incomplete grids cannot be ignored. For supported paired cases, setup amortization, workload size and mapping policy materially change measured costs. In particular, the $n = 256$, $d = 3$ CV case changes backend preference between a single candidate and a full grid. These observations motivate criterion-specific and reliability-aware computational choices, not a universally fastest implementation or a new smoothing estimator.

Declarations

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Ethics and consent. Synthetic numerical experiments only; author confirmation of submission-specific declarations is pending.
Code and data. Source, selected-file provenance and derived data are archived as stated in the reproducibility section. Raw measurements remain in the author-owned Windows checkout and same-disk backup.
Author contributions and correspondence. To be confirmed and approved by both authors before submission. Affiliation and corresponding-author email are subject to confirmation.

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